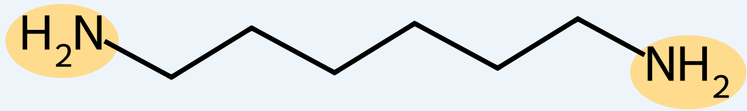
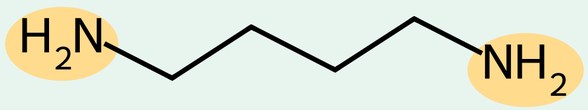
Native chemistry redirects the enzyme search

TARGET REACTION

Adipic acid + hexamethylenediamine (HMDA)



Six-carbon chain · two terminal amines

NATIVE NITROGEN DONOR		CHEMICAL RELEVANCE	SEARCH DECISION
GS	NH_3 Ammonia	These native reactions do not establish diamine acceptance.	**Deprioritize** these search scopes
GmaS	—NH_2 Methylamine		
PuuA	 Putrescine · four-carbon diamine	A native linear diamine motivates an HMDA lead.	**Expand the search** around the PuuA lead

Prioritize by native chemistry; target activity still needs testing.

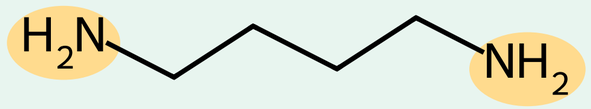
Native reactions: IUBMB EC 6.3.1.2 / 6.3.4.12; Kurihara et al., 2008. Actions: archived-run audit, events 51 and 122–125.

The chemical mismatch defines what to investigate next

PuuA is a lead. Can it accept both target substrates?

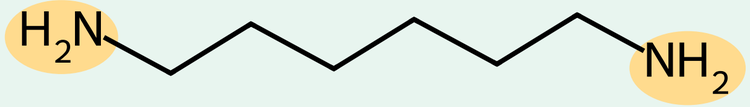
NITROGEN DONOR

Native



Putrescine

Target

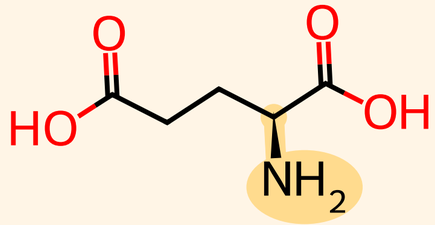


HMDA: longer by two carbons

Can the amine site accept the longer chain?

CARBOXYLATE DONOR

Native



L-glutamic acid

Target



Adipic acid: different donor chemistry

Can PuuA recognize and activate adipate?

PROPOSED NEXT STEP · Evaluate the actual adipate–HMDA pair

PuuA native substrates: Kurihara et al., JBC 2008, PMID 18495664. Neutral identity drawings; target activity is unestablished.